

# IEEE Sahrdaya Student Branch - Platform Report

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Create a comprehensive event ticketing and discovery platform featuring:

- **Zero-friction discovery** for all campus events
- **Society chair autonomy** for content management
- **Custom zero-free payment gateway** for event registrations
- **Digital student wallet** with QR-based ticketing + checkin support
- **Role-based access control** for secure operations

## Current Focus (Phase 1)

Build the foundation: public event discovery and society management interface with authentication infrastructure in place.

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## Phase 1: Implemented Features

### 1. Events Discovery Hub (/events)

**Status: Complete**

A centralized, dynamic event listing page showcasing all IEEE Sahrdaya events.

**Features:**

- **EventsHero component** - Highlights flagship upcoming events
- **Upcoming events grid** - Dynamic cards with event details (title, date, venue, hosting society, price)
- **Hall of Fame section** - Showcases completed past events
- **Society filtering** - Query parameter support (/events?society=[id])
- **Server-side rendering** - Optimized data fetching from Appwrite
- **Fully public** - No authentication required for viewing

**Data Source:** Appwrite events collection

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### 2. Societies Showcase (/societies)

**Status: Complete**

Interactive directory of all IEEE societies with inline editing capabilities.

**Public Features:**

- Grid layout displaying all 14 IEEE societies
- Society cards with logo, name, bio, and banner
- "View Events" link filtering to society-specific events

### Society Chair Features (Authenticated):

- **Conditional "Edit" button** - Appears only on chair's own society
- **Inline editing modal** - Update bio, banner, and logo
- **Real-time updates** - Changes reflect immediately via Appwrite
- **Document-level security** - Enforced by Appwrite Teams

**Data Source:** Appwrite [societies](#) collection

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## 3. Lazy Authentication System

**Status: Complete**

**Philosophy:** Public-first - authentication only when necessary.

### Features:

- **Google OAuth integration** via Appwrite
- **Contextual login prompts** - Appears only when user needs to:
  - Edit society details (if they're a chair)
  - Register for events (Phase 2)
- **Auth context provider** - Global state management
- **User profile dropdown** - Logout and profile management
- **Session persistence** - Maintained across pages

### Login Triggers:

- Clicking "Edit" on society card (chairs only)
- Attempting to register for events (Phase 2)

### No Login Required:

- Browsing homepage, events, or societies
  - Viewing event details
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## 4. Navigation & UI Enhancement

**Status: Complete**

- **Updated Navbar:**
  - Links to [/events](#) and [/societies](#)
  - Authentication status display
  - User profile dropdown with logout
  - Active page highlighting
- **Enhanced SocietyStrip:**
  - Now fetches from Appwrite (previously hardcoded)
  - Clickable logos navigate to [/societies](#)

- Loading states
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## 5. Backend Infrastructure

### Status: Complete

- **Appwrite client configuration** ([lib/appwrite.ts](#))
  - **Auth context provider** ([contexts/AuthContext.tsx](#))
  - **TypeScript types** - Society, Event, User, TeamMembership
  - **Environment variables** setup
  - **OAuth callback handler** ([/auth/callback](#))
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## Phase 2: Planned Features

### 1. Event Registration System

#### Priority: High

- Event detail page ([/event/\[id\]](#))
  - User profile collection (name, phone, department, year, semester)
  - Registration form with profile reuse
  - Capacity management and registration deadlines
  - Registration status tracking
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### 2. Admin Dashboard ([/admin](#))

#### Priority: High

#### Complete Event Management Platform:

- **Dashboard Overview:**
  - Total registrations, revenue, active events
  - Quick stats and recent activity
  - Society performance metrics
- **Event Management:**
  - Create upcoming events with registration
  - View all events across societies
  - Event analytics (registrations, capacity, revenue)
  - Registered students list with full data
  - Export attendee lists to CSV
  - Bulk email capabilities
- **Student Database:**
  - All registered students across platform
  - Profile data and participation history

- Search, filter, and export

- **Analytics & Reports:**

- Registration trends
  - Revenue by event/society
  - Participation by department/year
  - Capacity utilization metrics
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### 3. Enhanced Events Page (/events)

#### Three-tab interface for logged-in users:

- **Tab 1: All Events** - Browse and register
- **Tab 2: My Tickets** - Active event tickets with QR codes
- **Tab 3: Past Events** - Participation history

#### Features:

- Registration button on event cards
  - QR code ticket generation
  - Download ticket functionality
  - Add to calendar integration
  - QR Checkin support
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### 4. Society Chair Dashboard

#### Enhanced society management:

- Add/edit/delete past events archive
  - View events analytics
  - Registration data for their events
  - Society performance metrics
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### 5. ZeroGate Payment Engine

**Priority:** Medium

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### 6. Digital Student Wallet (/wallet)

**Priority:** Medium

- Active tickets with QR codes
  - Past event history
  - Download ticket PDFs
  - Participation certificates
  - Transaction history
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## 7. QR Check-in System

**Priority:** Low

- QR scanner for society chairs
- Real-time attendance tracking
- Check-in validation
- Duplicate entry prevention
- Entry analytics

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## Technical Architecture

### Technology Stack

**Frontend:**

- **Framework:** Next.js 15 (App Router)
- **Language:** TypeScript 5.8
- **Styling:** Tailwind CSS 3.4
- **Animations:** Framer Motion 12.34
- **Icons:** Lucide React 0.563
- **Fonts:** Press Start 2P (pixel), Inter (body)

**Backend:**

- **BaaS:** Appwrite 1.8.0 (Database, Auth, Teams, Storage)
- **Authentication:** Google OAuth via Appwrite
- **State Management:** React Context API

**Deployment:**

- **Hosting:** Vercel
- **DNS:** Cloudflare

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### Database Schema (Appwrite)

**societies** Collection

```
{
  name: string           // Society name
  slug: string           // URL-safe identifier (e.g., "cs")
  bio: string            // Description
  logo_url: string       // Logo image path
  banner_url?: string    // Banner image (optional)
  created_at: DateTime   // Auto-generated
}
```

**Permissions:** Read: Any | Update: Team-based (chair\_[slug])

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## events Collection

```
{
  title: string           // Event name
  description?: string     // Full event details
  date: DateTime          // Event start date/time
  venue?: string          // Location
  price: number           // Registration fee (0 for free)
  banner_url?: string     // Event banner (9:16 aspect)
  society_id: string      // Foreign key to societies
  status: string          // draft | published | archived | completed
  max_capacity?: number   // Max registrations (0 = unlimited)
  registration_enabled: boolean
  registration_deadline?: DateTime
  created_by: string      // User ID
  registrations_count: number
  created_at: DateTime
}
```

**Permissions:** Read: Any | Create/Update: Team-based

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## users Collection (Phase 2)

```
{
  email: string           // From Google OAuth
  name: string            // From Google OAuth
  phone: string           // Collected
  department: string      // Collected
  year_of_study: string   // e.g., "2nd Year"
  semester: string        // e.g., "S4"
  ieee_reg_no?: string    // Optional
  student_id?: string     // Optional
  role: 'student' | 'chair' | 'admin'
  society_id?: string     // If chair/admin
  created_at: DateTime
  updated_at: DateTime
}
```

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## registrations Collection (Phase 2)

```
{
  user_id: string         // Links to users
  event_id: string        // Links to events
  ticket_id: string       // Unique QR identifier
}
```

```

registered_at: DateTime
checked_in: boolean
checked_in_at?: DateTime
payment_status: 'pending' | 'completed' | 'free'
payment_amount: number
}

```

## File Structure

```

ieee-sahrdaya/
├── public/
│   ├── Events/           # Event images
│   ├── Execom/           # Team member photos
│   └── Societies/        # Society logos
├── src/
│   ├── app/
│   │   ├── layout.tsx    # Root layout with AuthProvider
│   │   ├── page.tsx      # Homepage
│   │   ├── auth/callback/ # OAuth handler
│   │   ├── events/page.tsx # Events discovery hub
│   │   ├── societies/page.tsx # Societies showcase
│   │   └── full-execom/   # Full team page
│   ├── components/
│   │   ├── EventsHero.tsx # Events page hero section
│   │   ├── EventCard.tsx  # Reusable event card
│   │   ├── SocietyCard.tsx # Society card with edit
│   │   ├── EditSocietyModal.tsx # Society editing interface
│   │   ├── LoginModal.tsx  # Google OAuth modal
│   │   ├── Navbar.tsx     # Navigation with auth
│   │   └── ...             # Other components
│   ├── contexts/
│   │   └── AuthContext.tsx # Auth state management
│   ├── lib/
│   │   └── appwrite.ts     # Appwrite client config
│   └── types.ts            # TypeScript definitions
├── .env.local              # Environment variables
├── next.config.mjs         # Next.js configuration
├── tailwind.config.ts     # Tailwind configuration
└── package.json           # Dependencies

```

## Security & Permissions

## Role-Based Access Control

### Students (Default)

- View all public content
- Register for events (Phase 2)
- Access personal wallet (Phase 2)

### Society Chairs

- All student permissions
- Edit ONLY their society's details
- Add/edit/delete their society's past events
- View their events' registrations (Phase 2)

### Admins

- Full platform access
- Create upcoming events for any society
- View all registrations and analytics
- Manage all societies
- Access admin dashboard

## Security Features

- **Google OAuth** - Verified student identities only
  - **Appwrite Teams** - Role-based document permissions
  - **Server-side validation** - All mutations checked
  - **Document-level security** - Chairs can only modify their resources
  - **Session management** - Secure token-based authentication
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### Phase 1 (Current)

- All pages publicly accessible without auth walls
- Dynamic events fetched from Appwrite
- Society chairs can edit own societies
- Permissions enforced at database level
- Login contextually triggered only when needed

### Phase 2 (Target)

- Students can register for events in < 1 minute
  - Society chairs can create events in < 2 minutes
  - Admins can export registrations in < 10 seconds
  - 100% registration success rate
  - Zero unauthorized access attempts
  - QR code generation < 1 second
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# Implementation Timeline

## Phase 1: Foundation

**Duration:** 4 weeks | **Status:** Complete

- Appwrite setup, database schema
- Authentication system, context providers
- Events page implementation
- Societies page with editing

## Phase 2: Registration & Admin

**Duration:** 7 weeks | **Status:** Planned

- Database collections (users, registrations)
- Events page tabs (All | Tickets | Past)
- Societies page inline editing enhancements
- Admin dashboard foundation
- Event analytics & student data
- Polish, testing, optimization

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## License & Credits

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Platform built with Next.js, Appwrite, and modern web technologies.

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## Resources

- **Appwrite Documentation:** <https://appwrite.io/docs>
- **Next.js Documentation:** <https://nextjs.org/docs>
- **Tailwind CSS:** <https://tailwindcss.com/docs>
- **Framer Motion:** <https://www.framer.com/motion>
- **Website:** <https://test.ieeesahrdaya.com/>
- **backend:** <https://backend.ieeesahrdaya.com/>

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*For questions or support, contact the IEEE Sahrdaya Technical Team.*